



Paul Leger, Ph. D. in Computer Science & Software Engineer

Software Engineer & Associate Professor
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Personal

Name: Paul Leger
Hometown: Coquimbo, Chile
Nationality: Chilean

Summary:

Google Scholar: CrqCdsgAAAAJ
ORCID: 0000-0003-0969-5139
SCOPUS: 36087550600
DBLP: <https://dblp.org/pid/78/7936.html>
Directors/Headships: 6 Positions
Indexed Journals: 37 Publications (32 WoS-JCR, 2 SCOPUS, and 3 Latindex)
Indexed Conferences: 20 Publications (SCOPUS)
Indexed Workshops: 7 Publications (SCOPUS)
Journal Editions: 6 Journals
First/Corresp Author: 24 (represents 37%)
Courses: 56 Courses (49 undergrad and 7 graduate)
Thesis Supervisions: 27 Theses (24 undergraduate and 3 graduate)
Relevant Software: 9 Developments (2 applications, 3 libraries, and 4 languages)
External Projects: 3 with funding



Education

Visiting Researcher, universities across the world:

- Centre for Development of Advanced Computing (C-DAC), India, 2024 (3 weeks).
- University of The Andes, Colombia, 2021 (3 weeks).
- Tokyo Institute of Technology, Japan, 2019 (6 months).
- University of Cauca, Colombia, 2018 (2 months).
- Shibaura Institute of Technology (x 2), Japan, 2013-2014 (4-2 months).
- École des mines de Nantes, France, 2010 (2 months).

Certifications: *a)* "Fundamentals of English Medium Instruction", University of Pennsylvania, 2023, *b)* "Teaching University", Highest level, Catholic University of the North, 2024.

Ph. D. in Computer Science, University of Chile, 2006 - 2012.

- Subject: Opening Up Trace-Based Mechanisms.
- Advisor: Éric Tanter (<https://pleiad.cl/etanter>).

Engineering degree in Computer Science, University of La Serena, 1999 - 2005 (July).

B.Sc. in Computing Science, University of La Serena, 1999 - 2005 (April).

Management Services

Director. Engineering School (<https://eic.ucn.cl>) at Universidad Católica del Norte (Chile). Management of a budget for a four-floor engineering school with 1,000+ students, guiding professors, and a teaching support unit, 2024 - 2026.

Director. Chilean Computer Science Society (SCCC), 2020 - 2022.

Deputy Director. Engineering Systems Doctorate (<https://dsi-en.ucn.cl>) at Universidad Católica del Norte (Chile), 2023 - 2024.

Head of Program Studies. 1) Civil Engineering in Computer Science and 2) Information Technology Engineering at the Universidad Católica del Norte (Chile), over 400 students, 2021 - 2022.

Pragmatics Lab. Director of an international research laboratory (<http://pragmatics-lab.com>), where professors and students from Japan, Colombia, and Chile collaborate, 2019 - present.

Software Development Engineer

As I enjoy working on software engineering, I occasionally work as a freelancer doing software development and as a team manager in software projects:

Macrosoftware. Consulting software engineer, 2020 - present.

G-Tech. Project Manager, a startup mobile application, 2022.

Autopista Central. Consulting engineer, 2017.

Regional Observatory (working market). Software project manager, implementation of an information system that allows users to analyze working market in a region of Chile, 2016 - 2017.

Autopista Central. Consulting engineer, 2011.

Software Development for Research

Research-driven software systems and developer tools, built with production-grade engineering practices where applicable (open-source repos, demos, and reusable libraries). Source: <https://pleger.cl/software>.

PL/SE research with AI: Implementation with many upgrades of my previous research in Programming Language and Software Engineering with Artificial Intelligence agents (Codex): AspectScript, Expressive Stateful Aspect (ESA-myPhD), DeloreanGo (adaptation from DeloreanJs), Sync/CC, and EMA-Kotlin (adaptation from EMA-js). <https://pleger.github.io/PL-SE-research-with-AI>, 2026. [ARTIFICIAL INTELLIGENCE].

SBABM: Java software to simulate buyers in e-commerce like Amazon, AliExpress, and Wish. An agent-based model to simulate the e-commerce behavior between agents and markets. <https://github.com/pragmatics-laboratory/SBABM>, 2023. [APPLICATION].

EMAJs: JavaScript library that implements an Expressive and Modular Activation mechanism for context-oriented programming. <https://github.com/pragmatics-laboratory/EMAJs>, 2022. [LANGUAGE].

DeloreanJS: A back-in-time debugger for JavaScript. This software is strongly developed by Felipe Ruiz (felipe.ruiz@alumnos.ucn.cl) and Guillermo Victorero (guillermo.victorero@alumnos.ucn.cl). You can try it on <https://pleger.cl/sites/deloreanjs>. <https://github.com/pragmatics-laboratory/deloreanjs>, 2019. [APPLICATION].

AspectScript: An aspect-oriented extension of JavaScript for expressive AOP, which integrates several state-of-the-art features related to higher-order programming and expressive scoping of aspects. <https://pleiad.cl/aspectscript>, 2010. [LANGUAGE].

FASOW: FASOW provides an Open Implementation for Agent-Based Models applied to e-WOM Marketing Campaigns. This software is strongly developed by Diego Nuñez (diego.nunez.trabajo@gmail.com). <https://github.com/pragmaticslaboratory/fasow>, 2024. [LIBRARY].

Matcher Cell: A JavaScript library to flexibly match patterns. You can try it on <https://pragmaticslaboratory.github.io/matcher-cells-study-cases>. <https://github.com/pragmaticslaboratory/match-cell-base>, 2021. [LANGUAGE].

Sync/CC (proof of concept): A library to address callback issues on JavaScript. You can try it on <https://pleger.cl/synccc>. <https://github.com/pragmaticslaboratory/synnccc>, 2019. [LIBRARY].

Academic Experience

Scientific Services

Funded Projects (*x* 4). Principal Researcher (ANID-2023), Principal Investigator (Chile), and Co-Researcher (Japanese government).

Organizer of International Events (*x* 4). JCC'20 (Chile), COP'20 (Germany), and MPSE'14/15 (USA).

Editor and Reviewer of JCR Journals (*x* 48). Editor (*x* 5) and Reviewer (*x* 43).

Chair of International Events (*x* 9). ICSME'23 (Colombia), LMN'22 (Chile), JCC'20 (Chile), MPSE'14/15 (USA), SAC-SE'17 (Morocco), and BIONETICS'15 (USA).

Program Committee Member (*x* 21). VMIL (Singapore), ICSME (various countries), SAC-SE (various countries), SBLP (Brazil), COP (Germany), VISSOFT (Chile), INFONOR (Chile), etc.

Evaluator (*x* 16). Research Projects and (International) External Master Reviewers.

References

Hidehiko Masuhara. Ph. D. in Computer Science. Full Professor & Dean. Department of Mathematical and Computing Science. Tokyo Institute of Technology, Japan (masuhara@acm.org). Webpage: <https://prg.is.titech.ac.jp/people/masuhara>.

Éric Tanter. Ph. D. in Computer Science. Full Professor. Computer Science Department (DCC), University of Chile, Chile (etanter@dcc.uchile.cl). Webpage: <https://pleiad.cl/people/etanter>.

Alexandre Bergel. Ph. D. in Computer Science. Computer Scientist. Rational AI (<https://relational.ai>), Switzerland (alexandre.bergel@me.com). Webpage: <https://bergel.eu>.